

SOLOMON GHANDI

DARE TO BE NAIVE

01
23



23 TRACKS

A JOURNEY THROUGH
SYSTEMS, GEOMETRY
AND THE UNIVERSE

SYNERGETICS

THE GEOMETRY OF THINKING



SCIENCE.
DESIGN.
INTELLIGENCE.
HUMANITY.
ONE SYSTEM.

PARENTAL
ADVISORY
EXPLICIT CONTENT

A CONCEPT ALBUM
BUILT AS A COURSE.
LIVE AS A PHILOSOPHY.



SOLOMON
GHANDI

SOLOMON

AI • CONSCIOUS HIP-HOP • SYSTEMS THINKING MUSIC

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THE PHILOSOPHY OF SOUND

Solomon Ghandi is an AI music artist created by a Ghanaian software developer with a vision to merge Artificial Intelligence, Music, and Systems Thinking.

Inspired by conscious hip-hop legends like [Lupe Fiasco](#) and the visionary concepts of [R. Buckminster Fuller](#), Ghandi's music transcends entertainment, serving as a metaphysical tool for global transformation.

The name "Ghandi" originated as a nickname during his time at Tema Secondary School in Ghana, symbolizing a commitment to intellectual and conscious leadership through art.

Solomon Yaw Adeklo is a software developer dedicated to shaping the future of conscious hip-hop music through AI.

[VIEW GITHUB](#)

Artist: Solomon Ghandi

Album: Synergetics: The Geometry of Thinking

Subtitle: Explorations in the Operational Mathematics of Universe

Label: Vector Equilibrium Soundworks

Format: 23-track concept album (2 LPs / digital)

SYNERGETICS: The Geometry of Thinking

A Concept Album by Solomon Ghandi

SYNERGETICS: The Geometry of Thinking is an ambitious educational concept album by **Solomon Ghandi** that merges conscious hip-hop, philosophy, systems science, engineering, geometry, mathematics, technology, and design into a single musical experience. More than an album, it is an intellectual journey that challenges listeners to rethink how they perceive themselves, nature, society, and the universe through the lens of relationships, patterns, and interconnected systems.

Designed as a musical curriculum rather than a conventional collection of songs, the album gradually introduces listeners to the language of systems thinking. Every track builds upon the previous one, transforming abstract scientific and philosophical ideas into engaging stories, memorable hooks, and thought-provoking verses. The result is a body of work that is equally at home in the classroom, the lecture hall, the engineering studio, and the headphones of curious minds seeking a deeper understanding of reality.

At its core, **SYNERGETICS: The Geometry of Thinking** explores one profound question:

How does the universe organize itself?

The album proposes that reality is best understood not as isolated objects but as dynamic systems connected through relationships, energy, structure, geometry, information, and continuous transformation. Mountains, rivers, cells, cities, ecosystems, technologies, civilizations, and galaxies are presented as expressions of universal patterns operating across different scales. By recognizing these recurring patterns, listeners are encouraged to think beyond disciplinary boundaries and develop a holistic understanding of the world.

Throughout the album, Solomon Ghandi examines morality as the foundation of sustainable civilizations, celebrates the curiosity of childhood as the beginning of scientific discovery, explores the hidden mathematical beauty of nature, and investigates the extraordinary complexity of human intelligence. As the journey progresses, listeners encounter concepts such as emergence, structure, systems, topology, spatial relationships, geometry, engineering design,

operational mathematics, modeling, and universal interconnectedness. Each composition transforms complex ideas into accessible narratives that inspire both critical thinking and creative imagination.

Rather than treating mathematics as a collection of equations or engineering as merely the construction of machines, the album presents them as languages for understanding reality itself. Geometry becomes the architecture of nature, systems become the framework of existence, and design becomes the process through which intelligence shapes the future. The project encourages listeners to recognize that every invention, every ecosystem, every civilization, and every scientific breakthrough emerges from relationships between interconnected components rather than isolated parts.

The album also challenges conventional education by demonstrating that science, philosophy, engineering, technology, ecology, and art are not separate disciplines but complementary ways of exploring the same universe. Through carefully crafted storytelling, lyrical metaphors, and conceptual progression, **SYNERGETICS: The Geometry of Thinking** seeks to cultivate systems thinkers capable of addressing the complex challenges of the twenty-first century.

Whether you are an engineer, scientist, architect, designer, programmer, mathematician, educator, student, philosopher, entrepreneur, or simply someone fascinated by the hidden order of the universe, this album offers an invitation to rediscover curiosity and embrace a more integrated way of thinking. It is a celebration of knowledge, creativity, innovation, cooperation, and humanity's endless pursuit of understanding.

More than music, **SYNERGETICS: The Geometry of Thinking** is a blueprint for intellectual exploration. It encourages listeners to move beyond memorizing facts and instead learn to recognize patterns, build meaningful connections, ask better questions, and design better solutions. Every song becomes a lesson, every verse becomes a thought experiment, and every chorus reinforces the idea that the universe is an interconnected masterpiece waiting to be understood.

Track Listing

1. Moral

Explores ethics, responsibility, and the role of moral principles in building resilient individuals, communities, and civilizations.

2. Child as Explorer

Celebrates curiosity, experimentation, imagination, and the fearless mindset that drives scientific discovery and lifelong learning.

3. Nature in a Corner

Reveals how universal principles can be observed in the smallest details of nature, from leaves and rivers to crystals and ecosystems.

4. Complex Humans

Examines humanity as a multidimensional system composed of biology, intelligence, culture, technology, emotion, and consciousness.

5. Metaphysics

Investigates the invisible foundations of reality, exploring ideas, perception, imagination, and the conceptual dimensions that influence the physical world.

6. Pattern

Introduces recurring structures and relationships that organize mathematics, biology, engineering, music, architecture, and civilization.

7. Space

Explores spatial relationships, dimensions, motion, proximity, and the geometry through which objects and systems interact.

8. Spherical Gravity

Examines planetary motion, attraction, balance, orbital systems, and the role of gravity in organizing the cosmos.

9. Wellspring

Celebrates creativity, imagination, innovation, and the continuous source of new ideas that drive human progress.

10. Humans In Universe

Places humanity within the broader context of Earth, nature, evolution, and the ever-expanding universe.

11. Synergy

Demonstrates how cooperation between individuals, ideas, and systems produces outcomes greater than the sum of their individual contributions.

12. Synergetics

Introduces an integrated philosophy of relationships, geometry, systems thinking, engineering, and universal design.

13. Universe

Explores the interconnected processes governing matter, energy, time, evolution, and cosmic organization across all scales.

14. System

Examines how components interact to create organized wholes, revealing the universal importance of structure, feedback, and interdependence.

15. Conceptuality

Investigates the power of ideas, abstraction, imagination, and conceptual thinking as the foundation of scientific and technological innovation.

16. Structure

Explores how organization, geometry, and relationships determine stability, function, adaptability, and resilience in natural and engineered systems.

17. Tensegrity

Examines how balanced tension and compression create efficient, stable, and resilient structures in engineering, biology, and architecture.

18. Operational Mathematics

Presents mathematics as a practical language for understanding transformation, measurement, prediction, relationships, and dynamic systems.

19. Modelability

Explores how maps, diagrams, simulations, prototypes, equations, and digital models enable humanity to understand and improve complex realities.

20. Omnitopology

Investigates universal connectivity, continuity, networks, boundaries, and the hidden architecture of relationships that unite seemingly separate systems.

21. Constant Zenith Projection

Explores new perspectives on geography, global mapping, planetary systems, and humanity's interconnected place on Earth.

22. Numerology

Reflects on humanity's enduring fascination with numbers, symbolism, mathematical order, and the search for meaning in patterns.

23. Evolution of Synergetics: A Portfolio

A grand finale that synthesizes the album's themes into a visionary reflection on the future of science, engineering, systems thinking, education, innovation, and humanity's continuing evolution toward a more integrated understanding of the universe.

"Every civilization is built twice—first in the imagination, then through design. Every discovery begins with curiosity. Every innovation begins with a pattern. Every revolution begins with a new way of thinking. *SYNERGETICS: The Geometry of Thinking* is an invitation to see the universe as one interconnected masterpiece—and to recognize your place within it." — Solomon Ghandi

TRACKS & LYRICS

1. Moral

Style: A cappella intro, single voice, cathedral reverb.

Concept: Derived from Fuller's "Moral of the Work" (<intro/moral.html>).

****[Verse]****

Dare to be naive.

Dare to be naive.

Do not refrain because the inventory seems complete—
local discovery breeds a complex of further discoveries,
heretofore unknown, previously unrecognized,
waiting in the wings of eternally regenerative Universe.

****[Chorus]****

God is the unknowable totality,
surprisingly unveiled,
synergetically inaugurating
entirely new, unpredictable ages.

****[Outro]****

Dare to be naive.

Dare to begin.

2. Child as Explorer

Style: Gentle fingerpicked guitar, building to layered percussion.

Concept: Scenario 100.01–100.63; the opening poem and the 32 unique geometrical features.

[Verse 1]

Universe to each must be all that is, including me.

Environment in turn must be all that is, excepting me.

I stand outdoors, I glance around,

I focus on a special case of generalized somethings—

a point-to-ability, a surface, a substance,

insideness and outsideness.

[Verse 2]

I break off a piece. I take a hammer to the rock.

Nature breaks big rocks into smaller rocks,

sand into grains,

until the minimum separable something reveals itself:

four corners, four faces, six edges,

twelve angles, concavity, convexity,

two poles of spinnability—

thirty-two unique geometrical features.

[Chorus]

Any four points in Universe

are most economically interrelated

by an ever-transforming tetrahedron.

I am the child as explorer.

I am the primitive, cosmic, equiwavelengthed frequency.

3. Nature in a Corner

Style: Tense, angular post-punk; staccato rhythms.

Concept: Scenario 260.00–269.07; “Getting Nature into a Corner,” Conceptual Minimum, Nothingness and Tunability.

[Verse 1]

Getting nature into a corner—

probe her with the operational procedure.

No solids, no straight lines, no continuous surfaces,

only the limit-case conditions

discovered by physical experience.

[Verse 2]

The conceptual minimum:

a tetrahedron of no size, timeless,

yet fractionable into differential polyhedra.

Observer as tetrasystem—

I am the tunability,

the omnioriented dial turning in.

[Chorus]

Topology of ins, outs, and interrelationships!

I corner the infinite

and find it is only a very large finite number
of very small, but finite multiples.

4. Complex Humans

Style: Cerebral art-rock; shifting time signatures, brass stabs.

Concept: Scenario 311.11–311.18; “Humans as Ultimate Complexities,” 92 regenerative elements, precession.

[Verse 1]

Assume a Universe with no substantive things—
only events, only awareness, only otherness.
Invent the ninety-two regenerative chemical elements,
their unique associative and repulsive behaviors,
billions of unique substances,
catalytically dissociating, re-associating.

[Verse 2]

Six equieconomical moves, four-dimensionally directed:
inwardly, outwardly, circumferentially around,
polarized involitionally-evolutionally inside-out.
Every event accomplished with six optional moves—
two, three, four, five radial zones outward,
or right back where you started.

[Chorus]

Complex humans!

Macromicro Universe-unfolding eventuation!

Physically irreversible, yet eternally integrated.

We cannot shrink back into the womb—
we can only evolve toward cosmic totality.

****[Bridge]****

It is doubtful that the million-light-year Andromeda nebula
has any interest in Republicans or Democrats,
communists or capitalists.

We are camouflaged beneath the blanket
of make-believe “reality.”

5. Metaphysics

Style: Ambient drone; spoken word over synthesizer pads.

Concept: Scenario 326.00–326.50; the “thinkable you” vs. the “touchable you,” weightless eternal principles.

****[Verse 1]****

You say you see me sitting here.

You see only pink face, hands, shoes, clothing.

You cannot see ***me***—

the thinking, abstract, metaphysical me.

****[Verse 2]****

What is important is the thinkable you,
the thinkable me,
the relatedness we have found,
the communications we have made.
When you hear Mozart on the radio,
you are intellect-to-intellect
with the livingly thinkable, eternal Mozart.

****[Chorus]****

Metaphysical! Utterly transcendent of the physical!
Weightless, omni-interaccommodative,
maximally synergetic, non-sensorial, eternal.
We recognize one another only as the touchable,
but we are the thinkable.

6. Pattern

Style: Minimalist techno; repeating modular sequences.

Concept: 505.50–505.83 / 606.01; “Mathematics is the science of structure and pattern in general.”

****[Verse 1]****

Mathematics is the science of structure and pattern in general.
Structure: a locally regenerative pattern integrity of Universe.
We cannot have a total structure of Universe.
Structure is inherently only local, inherently regenerative.

****[Verse 2]****

The necklace polygon, the necklace polyhedron—

tetrahedral, octahedral, icosahedral: stable.

Cubes and dodecahedra: unstable.

Only omnitriangulated systems are complete,

nonredundant, self-stabilizing.

****[Chorus]****

Pattern! Pattern integrity!

The rhythm of steps and heartbeats

monitors the memory-bank.

We remember patterns tactilely,

we run barefoot in the dark at ten miles per hour,

safe in the pattern.

7. Space

Style: Vast, reverberating ambient; celestial choirs.

Concept: 526.10–526.35; isotropic vector matrix, omnidirectionality, no “frames” of reference.

****[Verse 1]****

There are no continuous surfaces, no solids, no straight lines,

no infinitely extensible nonclosed-system planes.

The only infinity is the whole-fraction subdivisibility of wholes into parts,

the progressive halvings of the finitely closed circle.

****[Verse 2]****

The “frame” of reference is square—
imputed by two-dimensional language,
burdened by three-dimensional axes.
XYZ coordinates impose inept, exclusively rectilinear definings,
uncharacteristic of the omniwavilinear orbiting Universe reality.

****[Chorus]****

Space! Isotropic vector matrix!
Omnidirectional grid,
vertexes coinciding with sphere centers,
closest-packed, uniradius, omnispherical array.
Space is not empty; space is a system of relationships.

8. Spherical Gravity

Style: Heavy, circling bass; gravitational pull in the rhythm.

Concept: 646.10–646.22 / 251.04–251.07; gravity as spherically circumferential force, 12:1 advantage, 6:1 energy advantage.

****[Verse 1]****

Gravity is a spherically circumferential force.
Its effectiveness has a constant advantage ratio
of twelve to one over the radial inward mass-attraction.

The gravitational constant is always greater than the radiational—
exquisitely minute, but always operative.

****[Verse 2]****

Six to one energy advantage:
circumference chord-to-radius vectorial advantage.

The disintegrative forces of Universe
are effectively canceled out and embraced
by the integrative forces.

Contraction versus expansion—
the finite closure of the circumference wins.

****[Chorus]****

Spherical gravity!
Comprehensively embracing, circumferentially contractive.
Not the instinct to divide and conquer,
but the instinct to integrate and embrace.
Gravity is love with a 12:1 advantage.

9. Wellspring

Style: Solo piano, sparse, building to strings.

Source: Operational Mathematics (800.00), Moral, Preface.

****[Verse 1]****

Before the instrumentally informed history of musings,
before the microscope, the telescope, the electronic probing,

there was the child in tactile communication with the mother,
months inside the womb, visual muted, aural muted,
only the hand closing deftly around a finger,
opening again at the tension of departure.

****[Verse 2]****

The sequence is not accidental: touch is primary,
olfactory next, then aural, and only days later the eye.
Touch is yesterday, the thing you can get your hands on;
smell and taste and hearing are the now-set;
sight is the future-set, what only may be next,
the billionth-of-a-second lag between the page and the brain.

****[Verse 3]****

Blindfolded, mouth bound, ears closed—
only tactile sensing operative.
Sweeping out with arms extended, toe to fingertip,
a maximum spherical range of ten feet in diameter,
augmented by the ten-miles-per-hour dynamic
of running barefoot on the familiar camp path in the dark.

****[Verse 4]****

This is the wellspring: not the library, not the axiom,
but the operational procedure of the living.
The old life coaches the new life: eye, eye, eye;
but never points to the endocrine glands.
We begin with superficial identification,

and from this superficiality we must dive inward to the source.

****[Chorus]****

Wellspring!

All generalized principles are eternally operative,
waiting to be unveiled by freshly born, optimum-capacity human brains.

The wellspring never runs dry—
it only awaits the next drinker, the next naïve dare.

****[Outro]****

Dare to be naïve.

The source is in the sensing, the sweeping out,
the geometry of thinking before the geometry of proof.

10. Humans In Universe

Style: Orchestral folk; strings and acoustic guitar.

Source: 300.00 Universe, 311.10–311.18.

****[Verse 1]****

Universe is the comprehensive, historically synchronous,
integral-aggregate system embracing all separate integral-aggregate systems.
Nonsimultaneous, nonidentical, always complementary,
only partially overlapping macro-micro event sequences,
unitarily nonconceptual as of any one moment,
yet as an aggregate of finites, sum-totally finite.

****[Verse 2]****

Humans are each a special-case unfoldment-integrity
of the multi-alternated complex aggregate
of abstract, weightless, omni-interaccommodative,
maximally synergetic, non-sensorial, eternal principles.
We are physically irreversible, yet eternally integrated—
we cannot shrink and return into the womb.

****[Verse 3]****

Assume a Universe with no substantive things, only events.
Invent the ninety-two regenerative chemical elements,
their unique associative and repulsive behaviors,
billions of unique substances catalytically dissociating.
Invent precession and all the unique frequencies,
then watch the complexity multiply exponentially at a fourth-power rate.

****[Verse 4]****

Six equieconomical moves, four-dimensionally directed:
inwardly, outwardly, circumferentially around,
polarized involitionally-evolutionally inside-out.
Two, three, four, five radial zones outward from where you started,
or right back where you began—
six optional moves to close every event.

****[Chorus]****

Humans in Universe!

Varsity team members of the University of Universe!

Our brains are the storages that disclose
the previously undiscovered, unsuspectedly existent,
eternal generalized principles.

We can only evolve toward cosmic totality,
which in turn can only be evolvingly regenerated
through newborn humanity.

****[Bridge]****

It is doubtful that the million-light-year Andromeda nebula
has any interest in Republicans or Democrats,
communists or capitalists, or any terrestrial partisanship.

We are camouflaged beneath the blanket
of make-believe “reality” and economic me-firstings,
whose nonsensical preoccupation we call “today.”

11. Synergy

Style: Driving, celebratory rock anthem.

Source: 100.00 Synergy.

****[Verse 1]****

Synergy: the behavior of whole systems
unpredicted by the behavior of their parts.
Not the sum, not the product,
but the unexpected dividend of interaccommodation,

the comprehensive, coordinate system
employed by nature throughout all complementary transactions.

****[Verse 2]****

Universe to each must be all that is, including me.
Environment in turn must be all that is, excepting me.
The child stands outdoors, glances around, focuses:
a point-to-ability, a surface, a substance,
insideness and outsideness,
the minimum separable something revealing itself.

****[Verse 3]****

Four corners, four faces, six edges, twelve angles,
insideness, outsideness, concavity, convexity,
two poles of spinnability—
thirty-two unique geometrical features
in the minimum system of Universe.
Any four points are most economically interrelated
by an ever-transforming tetrahedron.

****[Verse 4]****

Four cosmically different tetrahedra coexist:
the tuned-in, at-presently-considered complex system;
the infra-tuned-in micro-tetra-nothingness;
the ultra-tuned-in macro-tetra-nothingness;
and the metaphysical, timeless-sizeless primitive concept.
Events are changes of interrelationships between a plurality of systems.

****[Verse 5]****

The whole, low-order, rational volume of the primitive
vector equilibrium, cube, octahedron, rhombic triacontahedron,
stated in tetravolumes: one, two-and-a-half, two-and-a-half,
three, four, five, six—
a hierarchy eternally invariable,
nonsimultaneously overlapping, only aggregately finite.

****[Chorus]****

Synergy!

The eternal generalized principles acting in concert,
unveiling the surprising payoff
of heretofore unknown problems
and their beautiful solutions.
Local discovery leads to a complex of further discoveries—
a complex of discoverabilities inherent in eternally regenerative Universe.

12. Synergetics

Style: Propulsive, intricate electronic; arpeggiated synths in 60-degree angles.

Source: 200.00 Synergetics.

****[Verse 1]****

Synergetics: the search for the omnirational,
comprehensive coordinate system employed by nature.

Not refuting Euclid, not refuting the hyperbolic or elliptic—
identifying the alternate freedoms of prime axiomatic assumption,
embracing all known mathematics,
eliminating the necessity for assigning physical qualities to irrational ghosts.

****[Verse 2]****

The radiolaria knew it before we did:
tetrahedron, octahedron, icosahedron—
microbiological structures based on nature's own design.
The picture was drawn by English scientists a century ago,
but I did not copy nature's structural patterns.
I began to explore structure in pure mathematical principle,
and the patterns emerged in pure principle.

****[Verse 3]****

Scientific entry into nuclear competence was accomplished
with the awkward irrational tools of CGS measurement
and the Cartesian XYZ ninety-degree coordinate system.
The awkwardness had to be corrected by Planck's constant.
Great computers have now relieved humans of the onerous tasks,
carrying the blind-flown pilgrimages on their silicon shoulders.

****[Verse 4]****

The geometrical model of energy is a symmetrical cluster of spheres,
each sphere a field of energy whose forces coordinate themselves,
shuntingly, pulsatively, momentarily asymmetrical,
never pausing at equilibrium.

The vector equilibrium is only a referential pattern
of conceptual relationships at which nature never pauses.

****[Chorus]****

Synergetics!

Angle and frequency added to Euler's inventory:
crossings, areas, lines—seven topological aspects,
plus frequency, always physically manifest.
Identify all experience in terms of only angle and frequency,
and you have nature's own geometry,
comprehensive because it describes instantaneously
both the internal and external limit relationships
of the sphere of energetic fields.

13. Universe

Style: Expansive, cinematic; choir and synthesizer swells.

Source: 300.00 Universe.

****[Verse 1]****

Universe is the aggregate of eternal generalized principles
whose nonunitarily conceptual scenario is unfoldingly manifest
in a variety of special-case, local, time-space transformative,
evolutionary events.

It is a dynamically synchronous scenario,
unitarily nonconceptual as of any one moment,

yet as an aggregate of finites, sum-totally finite.

****[Verse 2]****

It divides into six parts: the outside, the inside,
the prior, the subsequent, the system itself,
and all events occurring synchronously and coincidentally.
In addition to insideness and outsideness,
it is inherently concave and convex, complex, finite,
symmetrical or asymmetrical, a plurality of subsystems.

****[Verse 3]****

Humans are the ultimate complexities:
macromicro Universe-unfolding eventuation,
physically irreversible yet eternally integrated.
We are each a special-case unfoldment-integrity
of abstract, weightless, omni-interaccommodative,
maximally synergetic, eternal principles.

****[Verse 4]****

The scenario is nonsimultaneous, nonidentical,
only overlappingly occurring,
separate, differently enduring conceptual episodes,
their scenery, costumes, and character parts—
all special case, all temporal,
all coordinate to provide the ever-aggregating finiteness
of the nonsimultaneous whole.

****[Chorus]****

Universe!

The comprehensive, historically synchronous, integral-aggregate system!

We are the scenario, we are the scenery,

we are the costumes and the character parts—

all finite, all complementary,

all marching toward the evolution of cosmic totality.

14. System

Style: Mechanical, precise industrial rhythm; clockwork percussion.

Source: 400.00 System.

****[Verse 1]****

A system is the first subdivision of Universe.

It divides all the Universe into six parts:

outside, inside, prior, subsequent,

the system itself, and the synchronous coincident.

In addition to inherent insideness and outsideness,

a system is inherently concave and convex, complex, and finite.

****[Verse 2]****

Oneness, twoness, threeness cannot constitute a system—

they inherently lack insideness and outsideness.

Twoness is wavilinear relatedness;

threeness is planar relatedness, inherently triangular.

At minimum, it takes four triangular planes,
four vertexes, six common edges,
six wavelinear vectors connecting four event foci
to constitute prime withinness and withoutness.

****[Verse 3]****

The minimum system consists of two types of tetrahedra:
the mini-symmetric, the regular tetrahedron,
which cannot fill allspace alone,
and the mini-asymmetric, the irregular,
with greater enveloping strength per unit of contained phenomena.
The regular must compound with the octahedron,
twice the volume-to-surface ratio, to fill the void.

****[Verse 4]****

But two special-case minimum asymmetric tetrahedra—
the A Quanta Modules and the B Quanta Modules—
have equal volume and may be face-compounded
to fill allspace, uniquely the highest common volumetric multiple.
Two A Modules and one B Module form the MITE:
the prime minimum rational structural system,
the mathematically demonstrable microlimit of rational fractionation.

****[Chorus]****

System!

One-eighth the regular tetrahedron,
one-thirty-second the regular octahedron,

one-one-hundred-sixtieth the vector equilibrium of zero frequency,
one-one-thousand-two-hundred-eightieth the vector equilibrium of frequency two.
One thousand two hundred eighty A or B Modules
in the minimum realizable nuclear equilibrium—
and 1280 equals two to the eighth times five.
System is inherently local, inherently regenerative,
inherently the pattern of self-stabilization.

15. Conceptuality

Style: Dissonant, questioning; free jazz interludes.

Source: 500.00 Conceptuality.

[Verse 1]

The difference between synergetics and conventional mathematics
is that it is derived from experience
and is always considerate of experience.
Conventional mathematics is based upon axioms
imaginatively conceived, inconsiderate of information
harvested through microscopes, telescopes,
and electronic probings into the nonsensorially tunable ranges.

[Verse 2]

Solids, straight lines, continuous surfaces, and infinity
seemed imaginatively obvious, axiomatic.
Physics has discovered none of the foregoing

to be experimentally demonstrable.

The imaginary abstraction was so logical, so valid,
so obviously nonsubstantial in the preinstrumental history
that the mathematician assumed abstraction
to be systemic conceptuality absolutely devoid of experience.
He began with oversight.

****[Verse 3]****

Whereas experience is information and information is experience,
the only way we know that we are, that we are alive in Universe,
is through information apprehended by our own sensorial faculties.
We hear, see, taste, smell, and touch-feel.
The old life points to the eye, the mouth, the ear,
coaching the new life in superficial identification,
never referring to the endocrine glands,
never suspecting the depth of the operational procedure.

****[Verse 4]****

There are no dimensionless points, no continuous lines,
no infinitely extensible nonclosed-system planes.
The only infinity humanity has discovered experimentally
is the whole-fraction subdivisibility of wholes into parts,
the progressive halvings that divide the finitely closed circle
into ever smaller, central-angle-expressed arc increments.
Conceptuality must begin with the sensorially apprehended event,
not with the ghostly a priori.

****[Chorus]****

Conceptuality!

Not the abstraction devoid of experience,
but the operational mathematics of the living.

Experience first, theorem second.

The child knows before the proof.

The pattern is assumed significant unless proven otherwise.

16. Structure

Style: Sturdy, rhythmic; emphasis on angular, interlocking guitar riffs.

Source: 600.00 Structure.

****[Verse 1]****

Stability means angular invariability.

Inherent means behavior principles reliably operative
always and anywhere in Universe.

Regenerative means local energy-pattern conservation.

Constellar means an aggregation of enduring, cosmically isolated,
locally co-occurring events dynamically maintaining their interpositioning:
the Big Dipper, Orion, granite, cheese, flesh, atomic nuclei.

****[Verse 2]****

Mathematics is the science of structure and pattern in general.

Structure is a locally regenerative pattern integrity of Universe.

We cannot have a total structure of Universe.

Structure is inherently only local and inherently regenerative,
the triangle not only the unique pattern-self-stabilizing multienergied complex,
but accomplishing pattern stabilization at minimum effort,
coinciding with science's discovery of the omni-minimum-effort behavior
of all physical Universe.

****[Verse 3]****

In every triangle, each corner angle tension connector
serves as the common interfulcrum of two push-pull rigid lever arms.
Each pair of tubular necklace sides represents levers
whose maximum-advantage ends are seized by the third rigid side,
whose rigidity is imposed by its command of the two lever arm ends
upon the otherwise flexible opposite angle.
Thus the triangle stabilizes its opposite angle with minimum effort.

****[Verse 4]****

Tetrahedral, octahedral, and icosahedral necklace structures are stable.
Necklace cubes, rhombic dodecahedra, pentadodecahedra,
vector equilibria, and tetrakaidecahedra are unstable.
Only necklace-omnitrangulated, multifrequency geodesic spheres are stable,
based entirely on omnitrangulated tetra-, octa-, and icosahedral systems.

****[Verse 5]****

The number of vertexes of the omnitrangulated spherical structures
corresponds exactly with the number of external layer spheres
of closest-packed unit radius spherical agglomeration:
tetrahedra: two F squared plus two;

octahedra: four F^2 plus two;

icosahedra: ten F^2 plus two.

Only these three structural systems are complete, nonredundant, self-stabilizing.

****[Chorus]****

Structure!

Angular invariability at minimum effort!

The local regenerative pattern integrity of Universe,

forever rebuilding itself from the triangle up,

from the tetrahedron out,

from the geodesic sphere around.

17. Tensegrity

**Style:* Floating, disorienting; continuous drones with sudden percussive “islands.”

**Source:* 700.00 Tensegrity.

****[Verse 1]****

Continuous tension, discontinuous compression—

islanded staves of compression in dynamic stability.

The barrel, when in good material condition,

proves structurally stable because the internal load

distributes any local shock to all the enclosing surface,

thence to the finitely closed steel circle bands.

****[Verse 2]****

Whether pressure is exerted from outside or inside,
the result is always an outward thrust of the staves
against the tension members, whose finite closure
and cross-sectional strength ultimately absorb all working loads.
The vertical forces of gravity are translated precessionally
into horizontally outward buckling and torque stresses.

****[Verse 3]****

Molecules cannot be stacked up inside the sphere
in parallel or lesser-circle latitude planes.
Everything in motion travels in the direction of least resistance,
wherefore the great circle's inherent polar symmetries
must impose polar order—
an order hidden from the observer only by articulative velocities
that transcend the human optically tunable spectrum range.

****[Verse 4]****

Brouwer's theorem shows that when points are stirred randomly on a plane,
it can be proved mathematically that when the stirring stops,
one of the points was always at the center of the total stirring,
never disturbed in respect to all the others.
Any plane suitable for stirring must be part of a system
with an obverse surface polarly opposite,
producing poles in any bestirred complex system.

****[Verse 5]****

Push inward on two diametrically opposite points of a tensegrity sphere,

the whole sphere contracts symmetrically.

Pull outward on two islanded compression members,

the whole sphere precessionally and symmetrically expands.

Precession is not an exclusively single-plane, ninety-degree reorientation—
it operates omnidirectionally, as do all electromagnetic wave phenomena.

****[Chorus]****

Tensegrity!

Finitely closed circuits of continuous tension

restraining discontinuous, contiguously islanded staves of compression.

The barrel, the dome, the molecule, the cell—

all speaking the same structural language

of tension and compression in eternal negotiation.

18. Operational Mathematics

Style: Polyrhythmic, layered; each verse in a different “sense” texture.

Source: 800.00 Operational Mathematics.

****[Verse 1: Tactile]****

Blindfolded, mouth bound, ears closed.

Only tactile sensing operative.

Sweeping out with arms extended, reaching before shifting base,

prospecting with sensitive skin terminals like an insect with feelers.

Static range: one five-hundredth of a mile.

Dynamic: ten miles per hour on the familiar path in the dark.

****[Verse 2: Olfactory & Aural]****

Uncover the nostrils: static range one mile, dynamic four hundred.

Orange groves and pine trees at a mile offshore in still air.

Uncover the ears: static range one hundred miles, dynamic eleven hundred.

The boom follows the flash by a fraction—

the eye gets it faster than the ear,

and the child begins to realize there is lag in the rates of cognition.

****[Verse 3: Visual & The Lag]****

Uncover the eyes: static range six quintillion miles,

dynamic seven hundred million miles per hour.

The 186,000-miles-per-second speed of light is so fast

it was only recently measured, and it has no real meaning to us.

You insist you see the page, but you do not.

You have a brain-centered television set; the light bounces,

is scanned, goes back into the brain.

No one has ever seen outside themselves.

****[Verse 4: The Thinkable You]****

What is important is the thinkable you, the thinkable me,

the abstract metaphysical me, the relatedness we have found.

When you hear Mozart on the radio,

you are intellect-to-intellect with the eternal Mozart.

When you say “atom,” you are intellect-to-intellect

with the livingly thinkable Democritus.

You see only the touchable, nonthinking biological organism—

but we are the thinkable.

****[Chorus]****

Operational mathematics!

Information is experience. Experience is information.

Touch is yesterday; smell, taste, and hearing are now;
sight is the future.

Reality is eternally now, but human apprehending demonstrates
a large assortment of lags in rates of cognitions,
myriadly multivariied frequencies producing the illusion
of individually different awareness patterns.

19. Modelability

Style: Crystalline, precise; glass harmonica and digital synthesizers.

Source: 900.00 Modelability.

****[Verse 1]****

Modelability begins with the sphere-foldable great-circle set.

From one to another in the hierarchy of spinnable symmetries:

the three, four, six-sets of the vector equilibrium's twenty-five great-circle group,
and the six, ten, fifteen-sets of the icosahedron's thirty-one great-circle group.

The central angles of one often become the surface angles of the next,
while simultaneously some surface angles become the next sphere's central angles.

****[Verse 2]****

A triangle on the surface of the icosahedron
folds itself up, becomes a tetrahedron,
and plunges deeply down into the congruent central angles' void.
The Basic Triangle—one hundred twenty of which
are the lowest common denominator of a sphere—
makes a beautiful map, though its sinuses intrude into the continents.

****[Verse 3]****

Three steel straps joined by hollow stovepipe rivets,
rotatably journaled like scissors at the corners,
form an equilateral triangle of overlapping ends.
Gather the rods on one face side to a common point,
and the planes of rods rotate around their chordal axes,
forming a constant-length, equiedged, equiangled spherical triangle,
moving toward one another on the outer side,
away on the inner side.

****[Verse 4]****

The point to which all rod ends are gathered
becomes the center of a sphere on whose surface the three arcs occur,
arcs of great circles, for their planes pass through the center.
The sum of the corner angles adds to more than 180 degrees,
the spherical excess shared proportionately in each corner,
provided by the scissors-like angular increase
at each of the three journaled corners.

****[Verse 5]****

Release the rods from their common focus,
the spring-steel straps return to flatness,
all rods remain perpendicular to the steel bands throughout the transformation,
grouped in three separate and axially parallel planes.
What had been the external spherical zenith points remain in zenith—
the cartographic property of the constant zenith projection,
vital to the future of aerial and electromagnetic signal mapping,
communications satellites flying in fixed formation with Earth,
co-orbiting Sun at sixty thousand miles per hour.

****[Chorus]****

Modelability!

The mathematically demonstrable microlimit of rational fractionation,
both physically energetic and metaphysical,
a single universal geometrically discrete system-constant of quantation.
From the foldable triangle to the global map,
from the void to the vertex,
we model what nature has already proven.

20. Omnitopology

Style: Dense, swirling, overwhelming; information overload as aesthetic.

Source: 1000.00 Omnitopology.

****[Verse 1]****

The expression “frame of reference” is not only square

as imputed by two-dimensional language,
but also by its exclusive three-dimensional axes.
XYZ coordinates impose inept, exclusively rectilinear definings,
uncharacteristic of the omniwavilinear orbiting Universe reality.
Science has found no continuous surfaces, no solids, no straight lines,
no infinitely extensible nonclosed-system planes.

****[Verse 2]****

The only infinity humanity has discovered experimentally
is the whole-fraction subdivisibility of wholes into parts,
the progressive halvings that divide the finitely closed circle
into ever smaller, central-angle-expressed arc increments.
The spherical dimensions of tangent and angle frequencied intervals
can always be searchlighted “right on” all actual event tracery.

****[Verse 3]****

Self-deceiving misinterpretations were introduced by education,
by the momentum of tradition, by the persuasions of “common sense,”
by fear of questioning authority.
Macroscoped and microscoped data found no evidence
of dimensionless points, lines, planes, solids, or things—
no noun-designatable, thing-substantiated static entities.
Only verb-describable events: wavilinear trajectories, interferences,
resonant event fields.

****[Verse 4]****

Because there are spaces between closest-packed spheres,

energy can be imported syntropically all the way inward
to the prime nuclear domain, which thereafter can only be articulated outwardly—
ergo, as entropy.

The omnidirectional grid of the isotropic vector matrix,
whose vertexes always coincide with the sphere centers
of all closest-sphere packings, provides the new spherical reference system
that spontaneously accommodates the observer's omnirational accounting.

****[Verse 5]****

Precession operates omnidirectionally.

Push inward on any two diametrically opposite points,
the whole sphere contracts symmetrically.

Pull outward on two islanded compression members,
the whole sphere precessionally expands.

Waves can be reflectively focused and unidirectionally beamed,
but they are inherently omnidirectional,
as is all electromagnetic phenomena.

****[Chorus]****

Omnitopology!

Omnidirectional epistemology!

No nouns, only verbs. No things, only events.

The isotropic vector matrix accommodates
all intertransformative transactions
with omnivectorial, omnispherical, uniradius clarity.

21. Constant Zenith Projection

Style: Slow, majestic drone; navigational, celestial.

Source: 1100.00 Constant Zenith Projection.

[Verse 1]

Three steel straps, their ends joined by hollow stovepipe rivets,
perpendicularly inserted through registered terminal holes,
journaled pivotally together like a pair of scissors.

The three straps form an equilateral triangle
of overlapping and rotatably journaled ends,
the foundation of the transformational projection model.

[Verse 2]

Gather all the rods on one face side to a common point,
and each of the three spring-steel-strap sets yields in separate arcs.
The three planes of rods rotate around their chordal axes,
forming a constant-length, equiedged, equiangled spherical triangle.
The outer sections move toward one another;
the inner sections rotate away from one another.
The common focus becomes the center of a sphere.

[Verse 3]

The arcs are arcs of great circles, for their planes pass through the center.
The sums of the corner angles add to more than 180 degrees,
the spherical excess shared proportionately in each corner,
provided by the scissors-like angular increase at the three corners.

As the rods lengthen, the implied sphere's radius grows,
the central angles decrease,
the triangle approaches the pole of most extreme transformation.

****[Verse 4]****

Release the rods from the common focus,
the straps return to flatness,
all rods continue in perpendicularity to the steel bands,
grouped in three separate axially parallel planes.
The external spherical zenith points remain in zenith
in respect to each rod's point of penetration through the flat surface.
This is the cartographic property of the transformational projection.

****[Verse 5]****

The axis runs between the two poles of most extreme transformation,
through all the transforming triangular centers,
from the never-attained minimum-spherical-excess polar triangle
on the ever-enlarging sphere,
reversing toward the largest equatorial hemisphere phase,
then decreasing toward the never-attained smallest triangle
as the never-attained cosmically largest sphere expands.
Remember: the complementary triangle around the other pole
gets correspondingly larger.

****[Chorus]****

Constant zenith projection!
Rods remain in zenith throughout the transformation.

Satellites flying in fixed formation with Earth,
co-orbiting Sun at sixty thousand miles per hour,
mapping the world-surface-unified triangulation,
coordinating the spherical data of a planet in motion.

22. Numerology

Style: Hypnotic, incantatory; polyrhythmic chants, abacus percussion.

Source: 1200.00 Numerology.

[Verse 1]

Ancient numerologists developed tantalizingly logical theories:

congruence in modulo eight, modulo ten, modulo twelve.

Modulo ten: the convenience of ten fingers,

bendable, sequentially countable, memory-augmenting devices.

Modulo twelve: embracing the prime number three,

the triangle's stability, the father-mother-child.

But the one exception is the name for five,

derived directly from the human hand.

[Verse 2]

In the decimal system, zero through ten are cardinal.

But eleven and twelve—onze, douze, elf, zwoelf—

are cardinal in the duodecimal system,

their names retained even in decimal use.

After twelve, the ordinal naming begins:

thir-teen, four-teen, fif-teen—three plus ten, four plus ten.
Roman numerals were never numbers at all,
only tactile scratches matching visual sheep, one by one.

****[Verse 3]****

The abacus: thirteen columns of beads,
four below the bar, one above.
The empty column required a symbol—
the cypher, zero, the cipher.
Not until Algorismi in 1200 A.D.
did Europe learn how the cypher functioned calculatively,
moving tenness leftward, making possible
the Copernicuses, the Galileos, the Newtons.

****[Verse 4]****

Tobias Dantzig traced the names for numbers
in all the languages of Earth.
“One” starts with a vowel in every tongue—
eins, un, unus, yet, ahed.
“Two” starts with a consonant—
zwei, duo, dva, nee, tnayn.
The names were conceived by sailors,
carried by the Roaring Forties,
chanted in upside-downed longboats,
stored in rings on fingers, wrists, necks, and ears.

****[Verse 5]****

The Polynesians, naked on the sea,
invented congruence in modulo two:
yes-no, go-no go, the binary system
that now controls electronic circuitry.
Twenty spheres in a tetrahedron—no nucleus.
One hundred twenty spheres—no nucleus.
One hundred twenty: the least common denominator of the sphere,
the Magic Number in chemical-element isotopal abundance,
accommodating both decimal and duodecimal.
The numerologist integrates digits horizontally
until consolidated into one integer,
seeking the cosmic integrity of the prime.

****[Chorus]****

Numerology!

From the sailor's chant to the computer's byte,
from the abacus bead to the satellite's orbit,
we have always been counting the same Universe,
in modulo two, ten, twelve, and infinity.
The cypher was not nothing—
the cypher was the gate that broke the Dark Ages.

23. Evolution of Synergetics: A Portfolio

Style: Coda; solo voice, then full ensemble, then silence.

Source: Afterpiece / Preface / 203.09 / Moral.

****[Verse 1]****

In 1948, six drawings—
the first crystallizations of a lifetime's geometry.
Not copied from nature's structural patterns,
not arbitrary arrangements for superficial reasons,
but explored in pure mathematical principle,
out of which the patterns emerged,
developed themselves in pure principle,
realized as physical forms, applied to practical tasks.

****[Verse 2]****

The radiolaria had been photographed for a century,
but I did not see them until after I had produced the geodesic structures.
I did not copy nature; I did not make arbitrary arrangements.
I began to explore structure and develop it in pure mathematical principle,
and the patterns emerged.
The reappearance of tensegrity structures in scientists' findings
at various levels of inquiry confirms
the mathematical coordinating system employed by nature.

****[Verse 3]****

Pure coincidence—but excitingly valid coincidence.
The geometry was not in the books; it was in the principle, waiting.
Fuller is a Janus: grass-rooted in the past,
creating beacons for the very young.
Ivy-trained, rejecting academic discipline,

honored by the institution he transcended.

The danger of respectability lies in the loss of innocence,
in the guile that kills creativity.

****[Verse 4]****

Doing more with less.

There is no such thing as continuity and the infinitesimal:

with sufficient resolution we find only a very large, finite number
of very small, finite multiples, put together with very great ingenuity.

There always appears to be a structure.

Computer technology has moved in the direction of doing more with less.

What counts is the knowledge of how to put things together.

Knowledge is the tool of today and tomorrow.

****[Verse 5]****

Dare to be naive.

Please do not refrain because the inventory seems complete.

It is one of our most exciting discoveries

that local discovery leads to a complex of further discoveries.

Each age is characterized by its own astronomical myriads

of new special-case experiences and problems,

stored in freshly born optimum-capacity human brains,

which in turn disclose the presence

of heretofore undiscovered eternal generalized principles.

****[Chorus / Outro]****

Evolution of synergetics!

From the child's closed hand around the finger
to the geodesic dome enclosing a city,
from the tetrahedron of four corners
to the omnitopology of eternal regeneration—
we have done more with less.
We have dared to be naive.

(Silence. Then, barely audible:)

The portfolio is open.

The drawings are numbered.

The universe is finite but unbounded.

And the music—

the music is only the tuning in

and the tuning out.

(Silence)

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All lyrics derived from the text of R. Buckminster Fuller's Synergetics,

available online at rwgrayprojects.com/synergetics

Production Notes:

The album should be recorded with an awareness of “frequency as structure.” Where conventional rock music relies on 4/4 time (the “cubic” grid), Solomon Ghandi's rhythms should frequently pivot into groupings of 3, 6, 10, and 12—reflecting the closest-packing of spheres and the modular rhythm of the isotropic vector matrix. The transition from Track 8 (“Spherical Gravity”) into Track 9 (“Wellspring”) should be a single, seamless precessional shift, like a gyroscope reorienting at 90 degrees to its axis.

Liner Note Excerpt (to be printed on the inner sleeve):

“Synergetics is the integration of geometry and philosophy in a single conceptual system providing a common language and accounting for both the physical and the metaphysical. These songs are not explanations; they are operational procedures. Listen as you would explore—tactile first, then olfactory, then aural, then visual. The thinkable album is more important than the touchable vinyl.”

Recommended Entry Point:

As Bucky himself suggested, beginning listeners may wish to start at Track 23 and work backward, or begin with Track 2 (“Child as Explorer”) for the broad survey. The “Demass Model” is embedded in the silences between tracks.

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All lyrics derived from the text of R. Buckminster Fuller’s Synergetics*,
available online at rwgrayprojects.com/synergetics*

Dedicated to E. J. Applewhite, Arthur L. Loeb, and H. S. M. Coxeter